

## ME 625: Introduction to Turbulence and its Modeling

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| Course Code   | ME 625                                                                                                       |
| Course Name   | Introduction to Turbulence and its Modeling                                                                  |
| L-T-P-C       | 3-0-0-3                                                                                                      |
| Prerequisites | ME 210 Fluid Mechanics, IC 110 Engineering Mathematics/ ME 504 Numerical Methods for Engineering Computation |
| Intended for  | UG/PG                                                                                                        |
| Distribution  | Elective                                                                                                     |
| Approval      | 9th Senate (May 2016)                                                                                        |

### Course Contents

- **Introduction to Turbulence**  
Nature of turbulence, origin of turbulence, laminar and turbulent boundary layers, diffusion of turbulence, concept of eddy viscosity. [4 Lectures]
- **Statistics of Turbulence**  
Statistical aspects of turbulence, scales in turbulence, spectrum of turbulence, energy cascade in isotropic turbulence, Kolmogorov hypotheses. [6 Lectures]
- **Mathematical Theory of Turbulence**  
The Reynolds equation, Reynolds decomposition, equations for the mean flow, Reynolds stress, mixing length model, turbulent heat transfer, limitations of mixing length theory. [6 Lectures]
- **Dynamics of Turbulence**  
Dynamics of turbulence, Taylor microscale, Reynolds stress and vorticity, the vorticity equation. [4 Lectures]
- **Boundary-free and Wall-bounded Turbulence**  
Turbulent wakes, turbulent jets and mixing layers, turbulent flows in pipes and channels, experimental techniques for turbulence characteristics. [8 Lectures]
- **Introduction to Turbulence Modelling**  
Turbulence modelling and closure problem, algebraic models, modern variants of the mixing length model, one equation models,  $k - \epsilon$  and  $k - \omega$  models, Spalart-Allmaras turbulence model. [8 Lectures]
- **Numerical Techniques for Turbulence**  
Direct numerical simulations (DNS), large eddy simulations (LES) and Reynolds averaged Navier-Stokes (RANS) modelling techniques, spectral methods and particle based methods for turbulence. [8 Lectures]

### Text Books

1. Tennekes H. and Lumley J., **A first course in turbulence**, M.I.T. Press.
2. Tritton D.J., **Physical Fluid Dynamics**, Oxford University Press.
3. Davidson P.A., **Turbulence: An Introduction for Scientists and Engineers**, Oxford University Press.

### References

1. Pope S.B., **Turbulent flows**, Cambridge University Press, 2000
2. Townsend A.A., **The structure of turbulent shear flow**, Cambridge University Press, 1980.
3. Wilcox D.C., **Turbulence modeling for CFD**, DCW Industries, Incorporated, 1994.